plexus. Just behind the choroid plexus, the epithalamus forms a short stalk which ends in a rounded body called **pineal body** which secretes the hormone, **melatonin** which regulates sleep and wake cycle.

Thalamus is composed of grey mater which serves as a relay centre for impulses between the spinal cord, brain stem and cerebrum. Within the thalamus, information is sorted and edited and plays a key role inlearning and memory. It is a major coordinating centre for sensory and motor signalling.

Human brain is formed of a large number of parts like cerebrum, thalamus, hypothalamus, pons, cerebellum and medulla oblongata. Each part performs some specialized function and all the parts are essential for the survival of a person. Discuss the following statements :

a) Thalami are called relay centres of the brain.

b) Damage to medulla may cause the death of organism.

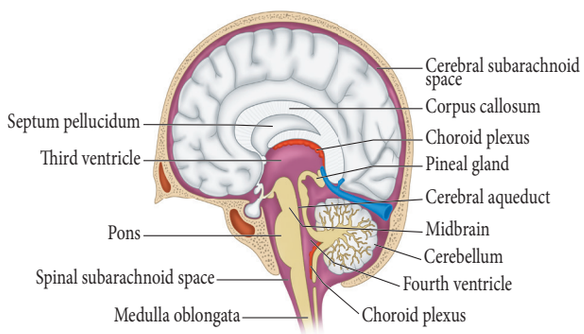


Figure 10.8 Mid sagittal section of brain

Hypothalamus forms the floor of the diencephalon. The downward extension of the hypothalamus, the **infundibulum** connects the hypothalamus with the pituitary gland. The hypothalamus contains a pair of small rounded body called **mammillary bodies** that are involved in olfactory reflexes and emotional responses to odour. Hypothalamus maintains homeostasis and has many centres which control the body temperature, urge for eating and drinking. It also contains a group of neurosecretory cells which secrete the hypothalamic hormones. Hypothalamus also acts as the **satiety centre.**

Depression is a functional deficiency of **seratonin** or **norepinephrine** or both. This disorder is characterized by a pervasive negative mood, loss of interest, an inability to experience pleasure and suicidal tendencies. Antidepressant drugs increase the available concentration of these neurotransmitters in the CNS. Hence depression is treatable.

Limbic system

The inner part of the cerebral hemisphere constitutes the limbic system. The main components of limbic system are **olfactory bulbs, cingulate gyrus, mammillary body, amygdala, hippocampus and hypothalamus.** The limbic system is called 'emotional brain' because it plays a primary role in the regulation of pleasure, pain, anger, fear, sexual feeling and affection. The hippocampus and amygdala also play a role in memory (Figure 10.9).

Brain stem is the part of the brain between the spinal cord and the diencephalon. It consists of mid brain, pons varolii and medulla oblongata (Figure 10.10).

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